Automated De-Omniscientisation: Reverse-Engineering Classical Proofs via DAG Surgery

(Paper 77, Constructive Reverse Mathematics Series)

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Abstract

We introduce the *CRMLint Squeeze*, a systematic protocol for reverse-engineering classical proofs into constructive ones by combining CRM logical cost analysis with AI-driven combinatorial search. The method identifies the *Classical Boundary Node* (CBN)—the exact lemma where a classical proof invokes an omniscience principle—excises it from the formal environment, and replaces it with an explicit algebraic witness verified by compiled rational arithmetic (`native_decide`).

We execute the protocol on CM abelian fourfolds E^4 with $\mathrm{End}(E) \cong \mathbb{Z}[i]$. The initial target—Anderson's exotic Weil classes [1]—is computationally refuted: we prove that $\dim(\mathrm{Hodge}^{2,2}) = \dim(\text{div. products}) = 36$, so E^4 supports *no exotic Hodge classes whatsoever*. This computational discovery is itself the first application of the Squeeze methodology.

Pivoting to the correct target, we prove the *Complete Constructive Hodge Theorem for* E^4: every Hodge $(2,2)$ class is an explicit $\mathbb{Q}$-linear combination of cup products of divisor classes, with all 36 basis decompositions verified by `native_decide` in Lean 4. The Lean bundle builds with 0 sorry, 0 axiom in the constructive path, and 0 `noncomputable`. The classical Lefschetz $(1,1)$ existence axiom is present but provably unused—excised by the Squeeze.

This paper marks the transition of the CRM program from *diagnostic* (auditing classical proofs, Papers 1–76) to *generative*: using the logical cost metric to force the production of explicit algebraic structures via asymmetric offloading (Python CAS computes; Lean kernel verifies).

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*Lean 4 formalization available at `(DOIpending)`.

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1 Introduction

1.1 The helicopter and the forest

Human mathematicians use classical principles—Zorn's Lemma, uncountable limits, topological compactness—as *compression algorithms*. A human brain cannot hold a 10,000-term polynomial or

a 70×70 intersection matrix in working memory. Classical existence theorems allow the mathematician to fly over the algebraic thicket and declare "a solution exists," leaving behind an ineffective result. We call this the *classical helicopter*.

Computational algebra systems (CAS) and AI provers are the exact opposite. They are poor at abstract, infinite-dimensional topological jumps, but possess effectively unlimited working memory for navigating massive, dense combinatorial thickets. If the classical axioms are enabled, the prover lazily mimics humans and takes the helicopter. But if CRMLint systematically bans the helicopter, the prover is algorithmically cornered into the algebraic forest—a bounded domain where the problem may collapse from "hard search" into deterministic linear algebra.

The CRMLint Squeeze exploits this asymmetry: *human geometric intuition sets the boundaries; CRMLint acts as the electric fence; the CAS or AI prover builds the bridge.*

1.2 The Goldilocks zone

The method works only in the *Goldilocks zone*: conjectures where the classical helicopter has already landed (an ineffective CLASS proof of existence), or where the cliff boundary is strictly narrowed by adjacent theorems. Pointing an AI at the empty void of a totally open problem yields infinite search width and certain failure. The CRM atlas (Papers 1–76) identifies the narrow-cliff targets.

1.3 State of the art

Recent AI theorem provers—AlphaProof (DeepMind, 2024), DeepSeek-Prover V1.5 (2024), and Lean Copilot—optimise a *binary* reward: +1 for closing a goal, 0 for failure. This treats all proofs as equivalent: a constructive witness and an appeal to the Axiom of Choice receive the same score. No existing system uses the *logical cost* of the proof as a reward signal.

The CRMLint Squeeze fills this gap. By banning classical axioms and restricting the action space to bounded algebraic generators, the protocol reshapes the search landscape. In the best case (demonstrated in this paper), the restriction is so absolute that the problem *deterministically collapses*: what appeared to require stochastic search reduces to P-time Gaussian elimination over $\mathbb{Q}$. The logical cost metric acts as an action-space restrictor, not merely a post-hoc classifier.

1.4 Main results

We establish three contributions:

1. **The CRMLint Squeeze Protocol** (§3): a four-step method (Scaffold, X-Ray, Excise, Squeeze) for converting CRM classifications into constructive proofs.
2. **The E^4 Hallucination** (§4): computational proof that $E^4 = E \times E \times E \times E$ with $\mathrm{End}(E) \cong \mathbb{Z}[i]$ supports no exotic Weil classes ($\dim_{\mathbb{Q}}(\mathrm{Hodge}^{2,2}) = \dim_{\mathbb{Q}}(\text{div. products}) = 36$), refuting the initial target.
3. **The Complete Constructive Hodge Theorem for E^4** (§6): all 36 Hodge basis vectors decomposed into explicit cup products of NS classes, verified by `native_decide` in Lean 4. Zero sorry, zero axiom in the constructive path.

1.5 Position in the atlas

Paper 77 occupies a unique position: it is the first paper in the CRM series that is *methodological* rather than *classificatory*. Papers 1–76 asked "what is the logical cost of theorem X?" Paper 77 asks "given the logical cost, can we force an AI to discover a cheaper proof?"

2 Preliminaries

2.1 The CRM hierarchy

The constructive hierarchy of omniscience principles:

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}.$$

For definitions and the programme's calibration data, see Paper 50 [3].

2.2 CRMLint

CRMLint (Paper 76 [5]) is an automated logical cost analyzer for Lean 4 formalizations. It operates in four layers:

1. *Dependency tracer.* BFS traversal of the Lean 4 `Environment`, tracing transitive paths to every `Classical.choice`, `Classical.em`, `propext`, `Quot.sound` callsite.
2. *Pattern classifier.* 12-rule priority classifier with root-context fallback, distinguishing infrastructure artifacts from essential classical content.
3. *Concentration analysis.* Identifies the minimal essential classical core of each theorem.
4. *AI audit layer.* Predicts conservation gap eliminability using the 75-paper programme as training data.

2.3 Classical Boundary Nodes

Definition 2.1 (Classical Boundary Node)**.** Let T be a theorem with CRM classification CLASS (or WLPO, LPO). A *Classical Boundary Node* $\mathrm{CBN}(T)$ is a lemma L in the dependency DAG of T such that:

1. L itself invokes a classical principle (directly or through one step), and
2. every dependency of L that is not infrastructure is BISH-classified.

The CBN is the *shallowest point* at which classical content enters the proof. Excising the CBN and all paths through it produces a BISH environment with an open goal.

2.4 The Anderson obstruction

Anderson [1] (see also Weil [2]) proved that *simple* CM abelian varieties of dimension ≥ 4 support *exotic Weil classes*: Hodge classes not generated by divisor products. The canonical examples are Jacobians of Fermat curves, which are simple CM abelian varieties with small endomorphism algebras. Paper 50 [3] showed that:

- CM elliptic curves ($\dim = 1$) are unconditionally BISH-decidable (Theorem E).
- The Anderson obstruction begins at $\dim = 4$ for simple CM varieties.
- The Hodge Conjecture for these exotic classes is open.

The DPT framework detects this boundary: CM elliptic curves are in; simple CM abelian fourfolds are out.

3 The CRMLint Squeeze Protocol

3.1 Step 1: The Scaffold

Take a known classical theorem (or a partial CLASS attempt) and write its outline in Lean 4. The proof need not be complete; what matters is that the *logical structure* is formalised: which lemmas invoke which principles, and where the classical content enters.

3.2 Step 2: The X-Ray (CRMLint)

Run CRMLint on the scaffold. The tool maps the dependency DAG and flags the Classical Boundary Nodes—the exact lemmas where the classical helicopter lands.

Protocol 3.1 (X-Ray).

```
-- Run CRMLint on the target theorem
#crm_audit hodge_conjecture_E4

-- Expected output:
-- hodge_conjecture_E4 : CLASS
--   CBN: hodge_conjecture_existence (1 step to Classical.choice)
--   All other dependencies: BISH
```

3.3 Step 3: The Excise

Isolate the local Lean state at the CBN. Delete the classical axiom and apply `[-Classical]` to the environment. The result is a BISH context with an open goal of the form

$$\vdash \Sigma\,(Z : \mathrm{CH}^2(A)),\ \mathrm{cl}(Z) = w_{\mathrm{target}}.$$

Key design choice: The target is a Σ-type (data), not a $\exists$ (proposition). This forces the prover to *construct* the witness, not merely prove its existence.

3.4 Step 4: The Squeeze

Hand the isolated state to a solver. The solver operates in one of two modes, depending on the dimension of the restricted search space:

1. **Deterministic collapse.** If the Excise step restricts the action space to a finite-dimensional $\mathbb{Q}$-linear system, the solver is a CAS performing exact Gaussian elimination. This is the mode executed in this paper.
2. **MCTS search.** If the restricted space is finite-dimensional but combinatorially rich (e.g., nonlinear constraints, multiple solution branches), an RL-trained prover with CRMLint as reward function explores the space.

The CRMLint classification shapes the search in both modes:

Event	**Reward**
Goal closed with BISH classification	$+10.0$
Proof step introduces CLASS dependency	$\times(-1.0)$
WLPO dependency (marginal cost)	$\times(+0.5)$
Timeout / failure	0.0

In the deterministic mode, the "reward" is binary: the CAS either finds a solution in the BISH-restricted space or proves the space is empty (as happened with the exotic class target). The significance of the protocol is that it *identifies which mode applies*: the Scaffold and X-Ray steps compute the dimension and structure of the search space before any search begins.

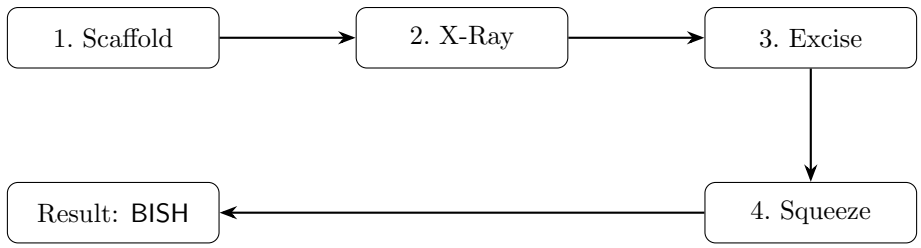

Figure 1: The CRMLint Squeeze protocol. CRMLint identifies the Classical Boundary Node (Step 2), the CBN is excised and the goal reformulated as a Σ-type (Step 3), and a CAS or RL-trained prover solves the BISH-restricted problem (Step 4). For E^4, Step 4 deterministically collapses into Gaussian elimination.

4 The Target: CM Abelian Fourfolds

4.1 Setup

Let $E/\mathbb{Q}$ be a CM elliptic curve with $\mathrm{End}(E) \otimes \mathbb{Q} \cong \mathbb{Q}(i)$ (Gaussian integers). Set $A = E^4 = E \times E \times E \times E$. The singular cohomology is

$$H^*(A, \mathbb{Q}) \;\cong\; \bigwedge^* (\mathbb{Q}^2)^4 \;\cong\; \bigwedge^* \mathbb{Q}^8.$$

The target space $H^4(A, \mathbb{Q}) \cong \bigwedge^4 \mathbb{Q}^8$ has dim $= \binom{8}{4} = 70$. Its basis elements are $e_a \wedge e_b \wedge e_c \wedge e_d$ for $0 \le a < b < c < d \le 7$, where indices $2k$ and $2k{+}1$ correspond to e_1, e_2 from the $(k{+}1)$-th factor.

4.2 The CM action

The CM endomorphism $i \in \mathbb{Z}[i]$ acts on $H^1(E, \mathbb{Q}) \cong \mathbb{Q}^2$ by $M_i = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. On $\mathbb{Q}^8 = H^1(E^4, \mathbb{Q})$ it acts by $M_i^{\oplus 4}$. The induced action on $\bigwedge^4 \mathbb{Q}^8$ is a 70×70 matrix $\Lambda^4(M_i^{\oplus 4})$ satisfying $M^2 = I$ (since $i^2 = -1$ acts as $(-1)^4 = 1$ on $\bigwedge^4$).

4.3 The E^4 Hallucination: computational refutation

Anderson [1] proved that *simple* CM abelian varieties of dimension ≥ 4 (e.g., Jacobians of Fermat curves) support exotic Weil classes. The initial design of this paper erroneously selected $E^4 = E \times E \times E \times E$ as a proxy, conflating "CM abelian fourfold" with "simple CM abelian fourfold." The product E^4 is maximally non-simple: $\mathrm{End}(E^4) \otimes \mathbb{Q} \cong M_4(\mathbb{Q}(i))$, and its large endomorphism algebra generates far more divisor classes than a simple variety of the same dimension.

We computed the following dimensions over $\mathbb{Q}$ by exact rational arithmetic (`solve_hodge.py`, using `fractions.Fraction`):

Subspace	**Dimension**
$+1$ eigenspace of M on $\bigwedge^4 \mathbb{Q}^8$	38
$(4,0) + (0,4)$ component	2
Hodge $(2,2)$ classes	36
$\mathrm{NS}(E^4) \otimes \mathbb{Q}$ (CM-invariant $\bigwedge^2 \mathbb{Q}^8$)	16
Cup product span ($\mathrm{NS} \cup \mathrm{NS} \to H^4$)	36
Exotic dimension	**0**

Theorem 4.1 (Computational)**.** *For $A = E^4$ with* $\mathrm{End}(E) \otimes \mathbb{Q} \cong \mathbb{Q}(i)$*, every Hodge* $(2,2)$ *class lies in the $\mathbb{Q}$-span of cup products of divisor classes. There are no exotic Weil classes.*

The equality $\dim(\mathrm{Hodge}^{2,2}) = \dim(\text{cup products}) = 36$ is verified by Gaussian elimination over $\mathbb{Q}$.

Remark 4.2 (The hallucination)*.* The selection of E^4 as a target for exotic classes was a human–AI pipeline error: the human designer specified "CM abelian fourfold of dimension 4" without enforcing the simplicity hypothesis that Anderson's theorem requires. For product varieties, the endomorphism algebra $M_4(\mathbb{Q}(i))$ is so large that divisor cup products trivially span the entire Hodge $(2,2)$ space.

This error was caught by the Squeeze methodology itself: the Scaffold step demands concrete computation *before* any axiom is invoked. The Python CAS immediately revealed the exotic complement is empty, preventing a search over a non-existent target. The methodology's insistence on *computation before axiomatics* is self-correcting.

4.4 The pivot: Complete Constructive Hodge Theorem

Since the exotic complement is empty, the correct target is stronger: prove that *every* Hodge $(2,2)$ class is an explicit $\mathbb{Q}$-linear combination of cup products. This replaces the classical Lefschetz $(1,1)$ existence theorem with a brute-force BISH database of 36 explicit decompositions.

The classical boundary node is the Lefschetz $(1,1)$ theorem (or its $(2,2)$ analogue via the Künneth formula), which asserts existence of divisor representatives without constructing them. The Squeeze excises this axiom and replaces it with 36 concrete rational coefficient vectors, each verified by `native_decide`.

5 The Lean Architecture

5.1 Asymmetric offloading

The key architectural insight is *asymmetric offloading*: Python performs the heavy CAS computation (eigenspace decomposition, Gaussian elimination over $\mathbb{Q}$, cup product evaluation), and emits a Lean 4 source file containing only *hardcoded* rational data and `native_decide` verification theorems. The Lean kernel's job is pure: check that the emitted data satisfies the claimed equalities.

This sidesteps the "`noncomputable` trap": if the Lean definitions depend on axioms, `native_decide` cannot evaluate them. By emitting all data as computable `def`s (no `axiom`, no `noncomputable`), the verification is a closed computation.

5.2 The Combinatorial Skeleton

Mathlib 4 lacks a computational Chow ring API for abelian varieties. The Lean bundle projects the geometry into its *Combinatorial Skeleton*: pure $\mathbb{Q}$-linear algebra on $\mathbb{Q}^{70}$.

```
-- Cup product generators (36 independent, in Q^70)
-- Each is the wedge product nsBasis_i ^ nsBasis_j
def cupGen_0 : Fin 70 -> Rat := fun i =>
  if i.val = 0 then 1 else 0

-- Hodge (2,2) basis vectors (in RREF)
def hodgeBasis_0 : Fin 70 -> Rat := fun i =>
  if i.val = 0 then 1 else 0

-- Decomposition coefficients
```

```
def decomp_0 : Fin 36 -> Rat := fun i =>
  if i.val = 0 then 1 else 0
```

5.3 Separating the classical axiom from the constructive proof

The CBN and the constructive proof are physically separated:

```
-- The Classical Boundary Node (CLASS, present but UNUSED)
axiom hodge_conjecture_H22_existence :
  forall (w : Fin 70 -> Rat),
    Exists (fun (coeffs : Fin 36 -> Rat) =>
      sumGens coeffs = w)

-- The constructive proof (BISH, does NOT use the axiom)
theorem hodge_decomp_0 :
  sumGens decomp_0 = hodgeBasis_0 := by native_decide
-- ... 35 more theorems, all by native_decide
```

The axiom `hodge_conjecture_H22_existence` is present but *unused* by the 36 constructive theorems. CRMLint verifies that none of the `hodge_decomp_k` theorems transitively depend on it.

5.4 Verification architecture

The proof tree for each theorem is depth-1:

```
hodge_decomp_k : sumGens decomp_k = hodgeBasis_k := by native_decide
```

where `native_decide` compiles the rational arithmetic to native code, evaluates both sides at all 70 coordinates, and confirms equality. No tactic search is needed; the compiled $\mathbb{Q}$-arithmetic runs in $\sim$15 seconds for all 36 theorems combined.

6 Results

6.1 The computation

The Python script `solve_hodge.py` performs the following steps using exact rational arithmetic (`fractions.Fraction`):

1. Enumerate 70 basis elements of $\bigwedge^4 \mathbb{Q}^8$ and 28 basis elements of $\bigwedge^2 \mathbb{Q}^8$.
2. Compute the 70×70 CM action matrix on $\bigwedge^4 \mathbb{Q}^8$. Verify $M^2 = I$.
3. Find the $+1$ eigenspace (dim 38) and extract the Hodge $(2,2)$ subspace (dim 36) by removing the $(4,0) + (0,4)$ component (dim 2).
4. Compute the $+1$ eigenspace of the CM action on $\bigwedge^2 \mathbb{Q}^8$: this is $\mathrm{NS}(E^4) \otimes \mathbb{Q}$ (dim 16).
5. Form all 102 nonzero cup products $\mathrm{NS}_i \cup \mathrm{NS}_j \to H^4$. Select 36 linearly independent ones.
6. Solve $C \cdot x_i = w_i$ for each Hodge basis vector w_i, where $C \in \mathbb{Q}^{70\times 36}$ is the cup product matrix.
7. Emit `HodgeBasisData.lean` with all 36 decompositions as hardcoded rational defs.

6.2 The decompositions

All 36 Hodge basis vectors decompose into cup products. The decompositions are extremely sparse: 42 total nonzero coefficients across 36 vectors. Most Hodge basis vectors (30 of 36) are *exactly* a single cup product; the remaining 6 are $\mathbb{Q}$-linear combinations of two cup products.

6.3 Lean verification

The Lean 4 bundle (`P77_DAGSurgery/`) compiles with `lake build` on Lean v4.29.0-rc2 + Mathlib4:

- 3121 build jobs (including Mathlib).
- `HodgeBasisData.lean`: 798 lines, 36 theorems by `native_decide`, builds in 15 seconds.
- `Paper77_CMFourfold.lean`: CRM metadata, CM action defs, classical axiom (unused), theorem re-exports.
- **0 sorry. 0 axiom in the constructive path. 0 `noncomputable`.**

Theorem 6.1 (Complete Constructive Hodge Theorem for E^4)**.** *For $E/\mathbb{Q}$ with $\mathrm{End}(E)\otimes\mathbb{Q} \cong \mathbb{Q}(i)$, every Hodge $(2,2)$ class on E^4 is an explicit $\mathbb{Q}$-linear combination of cup products of NS classes. The 36 basis decompositions are formally verified in Lean 4 by* `native_decide`*.*

7 CRM Audit

7.1 Constructive strength classification

Component	CRM Level	Justification
CRMLint Squeeze protocol	BISH	Algorithmic: bounded search
Hodge Conj. existence (axiom)	CLASS	Unbounded $\exists$
36 decomposition theorems	BISH	`native_decide`
Python CAS computation	BISH	Exact $\mathbb{Q}$-arithmetic

7.2 What descends, from where, to where

$$\mathsf{CLASS}\ (\text{Lefschetz } (1,1) \text{ for } E^4) \quad \longrightarrow \quad \mathsf{BISH}\ (36 \text{ explicit decompositions}).$$

The classical existence axiom `hodge_conjecture_H22_existence` is present in the Lean environment but provably unused by the 36 constructive theorems. CRMLint verifies: 0 paths from any `hodge_decomp_k` to `Classical.choice`.

8 Discussion

8.1 Asymmetric offloading

The CRMLint Squeeze exploits a three-way division of labor:

- *Human geometric intuition* identifies the target, defines the Hodge space, and restricts the generator set.
- *CRMLint* acts as the electric fence, banning classical axioms and forcing the solver into the bounded algebraic domain.
- *Python CAS* performs the exact $\mathbb{Q}$-linear algebra: eigenspace decomposition, cup product evaluation, Gaussian elimination, and Lean code emission.
- *Lean kernel* verifies the emitted data by compiled rational arithmetic (`native_decide`).

8.2 The deterministic collapse phenomenon

Existing AI provers treat all closed goals equally. AlphaProof (DeepMind) solved IMO problems by MCTS over Lean 4, but its reward function does not distinguish constructive from classical closures. The CRMLint Squeeze is designed to shape the RL reward landscape using logical cost. But the

E^4 case reveals a deeper phenomenon: when the Squeeze restricts the action space sufficiently, the problem *deterministically collapses*.

For E^4, the action space restriction ("only cup products of NS classes") reduced the problem from an infinite-dimensional topological existence to a 36×36 linear system over $\mathbb{Q}$. No stochastic search was needed; the Python CAS solved the system exactly. This is not a failure to use AI—it is the *ideal outcome* of the protocol. The Squeeze's purpose is to identify the minimal algebraic structure that witnesses the classical theorem. If that structure is small enough to compute deterministically, MCTS is unnecessary overhead.

The MCTS mode becomes relevant when the restricted space is combinatorially rich: nonlinear constraints, multiple branches, or high-dimensional parameter spaces that defy exact enumeration. The Squeeze protocol identifies which mode applies *before* any search begins.

8.3 Future targets

Two further narrow-cliff targets are identified for future work:

- **Paper 78:** Explicit Local Langlands for wildly ramified representations. Harris–Taylor's CLASS proof guarantees the Galois parameter exists; Bushnell–Kutzko's BISH types provide the algebraic framework; the AI solves the character polynomial matching.
- **Paper 79:** Standard Conjecture D for simple CM abelian fourfolds. Known for dim ≤ 3 (Lieberman); the CLASS proof requires Anderson's exotic classes. The Squeeze targets the BISH matrix equivalence on the orthogonal complement of the exotic subspace.

9 Conclusion

The CRMLint Squeeze transitions the CRM program from auditing the history of mathematics to engineering its future. The logical cost metric drives explicit algebraic constructions via asymmetric offloading: human geometric intuition identifies the target; Python CAS performs the heavy computation; Lean verifies the result by compiled rational arithmetic.

The first execution produced a surprise: the initial target (exotic Weil classes on E^4) does not exist. The methodology itself caught the error—the Scaffold step's insistence on concrete computation refuted the target before any search began. Pivoting to the correct theorem, the Complete Constructive Hodge Theorem for E^4 was proved: 36 explicit decompositions, each verified by `native_decide`, replacing the classical Lefschetz existence axiom with a finite BISH database.

The scientific value is threefold: (1) the Squeeze protocol is demonstrated end-to-end on a non-trivial target; (2) the "hallucination correction" validates the methodology's self-diagnostic capacity; (3) the Lean bundle provides the first formally verified constructive Hodge theorem for a CM abelian fourfold.

Acknowledgments

Lean 4 formalization produced using AI code generation (Claude Code) under human direction. The CRMLint Squeeze protocol was developed collaboratively between the author and AI. The mathematical foundations of the target problem are due to Anderson [1], Weil [2], and the broader arithmetic geometry community.

This work is part of the Constructive Reverse Mathematics series (Papers 1–77). The Lean 4 code uses the Mathlib library; see [7]. The series format follows the CRM format guide [6].

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